Day 18 | Friday

18

November 27, 2026

Perimeter, frame lengths and material totals

9:00 AM–10:15 AM • Math • 75 min

Circumference, arcs and perimeter consolidation

10:30 AM–12:00 PM • Math • 90 min

Week 5 • 2 scheduled blocks • 165 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

9:00 AM–10:15 AM • 75 min

Perimeter, frame lengths and material totals

9:00 AM–9:05 AM	**Start**	5
9:05 AM–9:10 AM	**Recall**	5
9:10 AM–9:25 AM	**Teach**	15
9:25 AM–9:40 AM	**Together**	15
9:40 AM–10:00 AM	**Practise**	20
10:00 AM–10:10 AM	**Check**	10
10:10 AM–10:15 AM	**Wrap up**	5

Math • Math Week 5, Lesson 6

10:30 AM–12:00 PM • 90 min

Circumference, arcs and perimeter consolidation

10:30 AM–10:35 AM	**Start**	5
10:35 AM–10:40 AM	**Recall**	5
10:40 AM–10:55 AM	**Teach**	15
10:55 AM–11:10 AM	**Together**	15
11:10 AM–11:45 AM	**Practise**	35
11:45 AM–11:55 AM	**Check**	10
11:55 AM–12:00 PM	**Wrap up**	5

Math • Math Week 5, Lesson 7

Break: 10:15 AM–10:30 AM • 15 minutes • excluded from teaching time

Perimeter, frame lengths and material totals

Today’s goal: Calculate a straight-sided perimeter and identify missing dimensions without counting internal lines.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Exercise Book 9A Q1–8 (p.85); 9B Q1 (p.87); 9C Q1–2 (p.89)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Recall and reconnect

Try to trace the outside of a sketch; contrast perimeter with area.

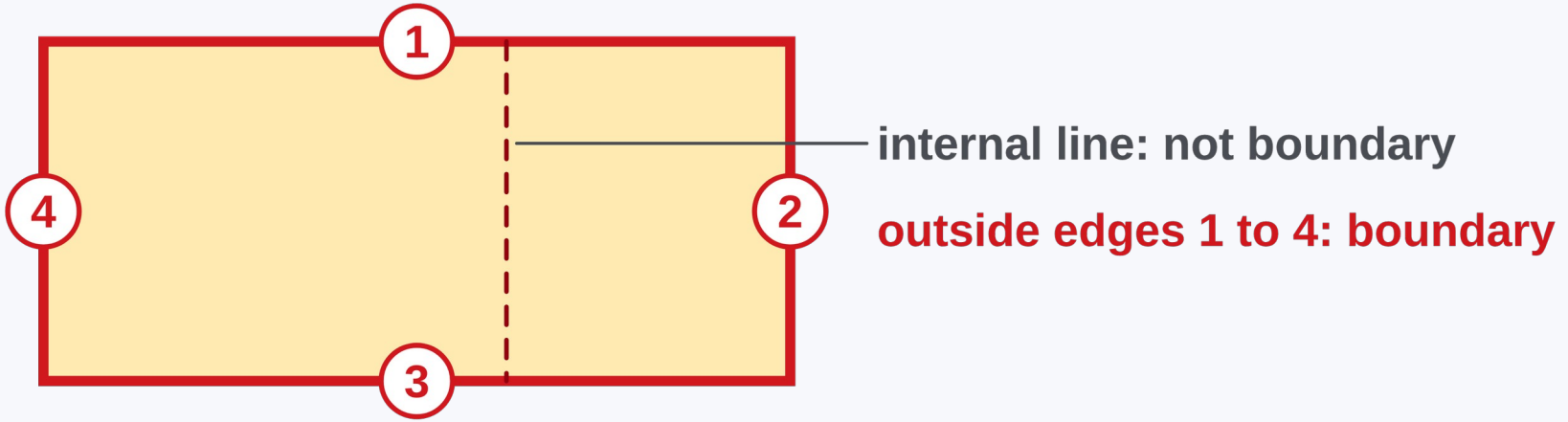
Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 9A Q1–8 (p.85); 9B Q1 (p.87); 9C Q1–2 (p.89)

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Trace the outside boundary

1. Perimeter is the total length around the outside of a shape.
2. Mark each boundary segment once before adding lengths.
3. An internal line or dashed height is not part of the boundary.
4. Use one length unit throughout and report a linear unit.



Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

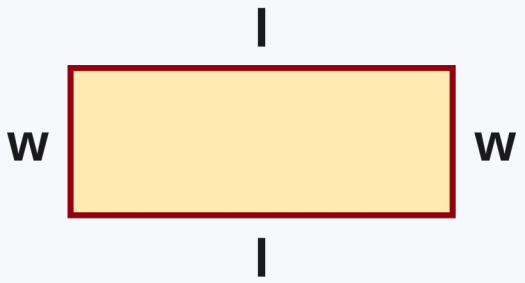
Find missing sides before adding

1 For a rectangle, opposite sides match.

2 Use P = 2(l + w) or add all four sides as a check.

3 Convert mixed feet and inches to one unit before calculating.

4 For other shapes, use only side relationships supported by the drawing.



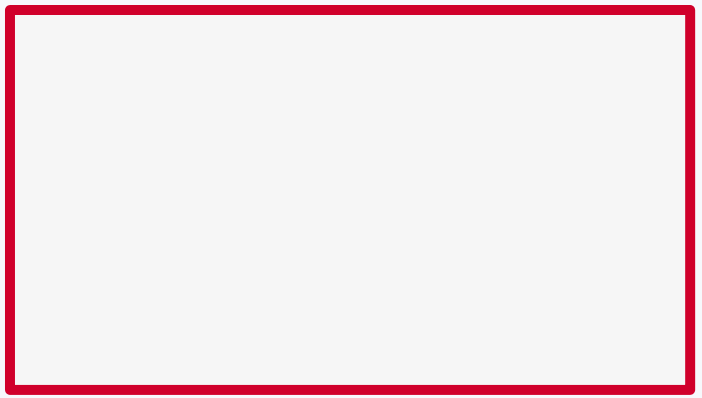
opposite sides match

l + w + l + w = 2(l + w)

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Trace the boundary before calculating

2 ft 6 in = 30 in



1 ft 9 in
= 21 in

P = 2(30 + 21)
= 102 in
= 8 ft 6 in

Check by adding:
30 + 21 + 30 + 21 = 102 in.

Together • 9A Q1,6–8 and 9C Q2.
Mark each boundary segment used. Compare addition with a formula.
Shape Canvas: rectangle and triangle; distinguish edges from height lines.

Textbook pp.220–221 • Exercise Book 9B Q1 p.87 • Diagram is schematic; read labels, not screen size

Textbook | pages 220–221

The **perimeter** is the distance all the way around the outside of a flat shape. Picture walking along every edge until you are back where you started. Add up every edge you walked.

PERIMETER

Rectangle: P = 2 × (l + w) **Square: P = 4 × s**

Triangle: P = a + b + c **Regular polygon: P = n × s**

l length, w width, s side, a, b and c the three sides, n the number of equal sides

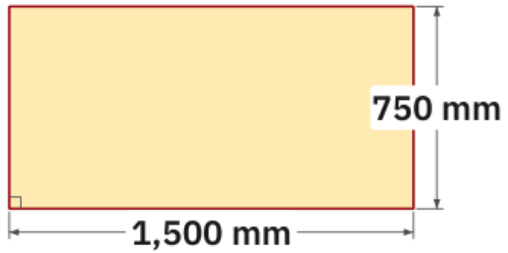
Textbook pp.220–221 • Perimeter — Outside boundary and formulas

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Textbook | page 220

Example 9.1 A rectangular guard frame

A guard frame is 1,500 mm long and 750 mm wide. How much flat bar goes around it?



SOLUTION

1. Two lengths and two widths: P = 2 × (l + w).
2. P = 2 × (1,500 + 750) = 2 × 2,250 = 4,500

Answer **4,500 mm, which is 4.5 m.**

Textbook p.220 • Example 9.1 — A rectangular guard frame

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Spot the mistake. Explain the correction.

A triangle shows a dashed height. Should it be added to the three sides for perimeter?

Discuss the method before checking the answer.

No. The height is an internal measuring line; trace and total the three outside edges.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Work together

Work Exercise Book 9A Q1,6–8 and Exercise Book 9C Q2. Mark every boundary segment used.

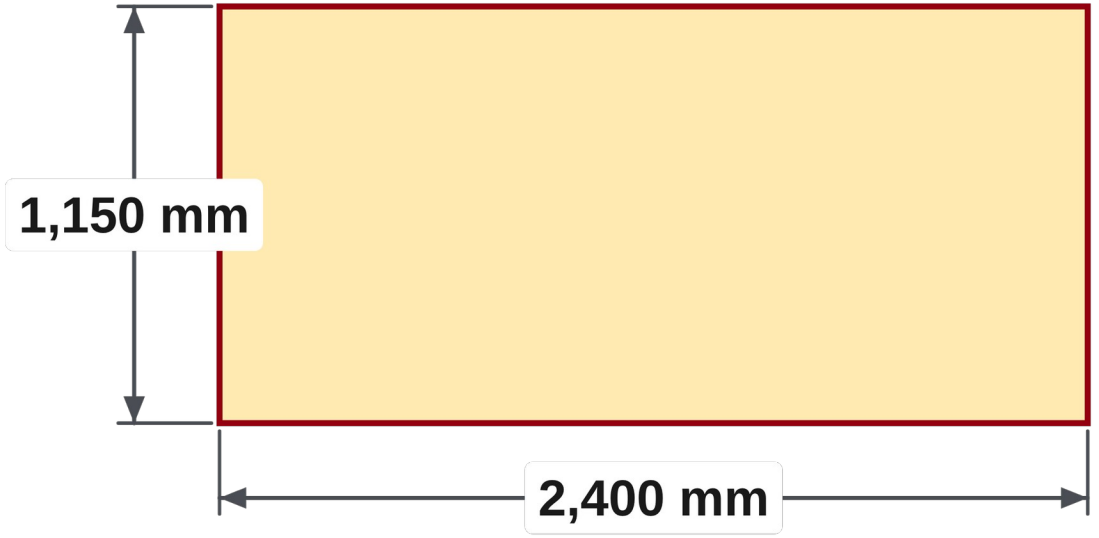
Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 9A Q1–8 (p.85); 9B Q1 (p.87); 9C Q1–2 (p.89)

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Guided practice | 9A Q1

Find the perimeter of a rectangle 2,400 mm by 1,150 mm.



Perimeter ?

Work with a partner. Take turns explaining the next step.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Compare your method | 9A Q1

Find the perimeter of a rectangle 2,400 mm by 1,150 mm.

CHECK TOGETHER

7,100 mm

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Your scheduled practice

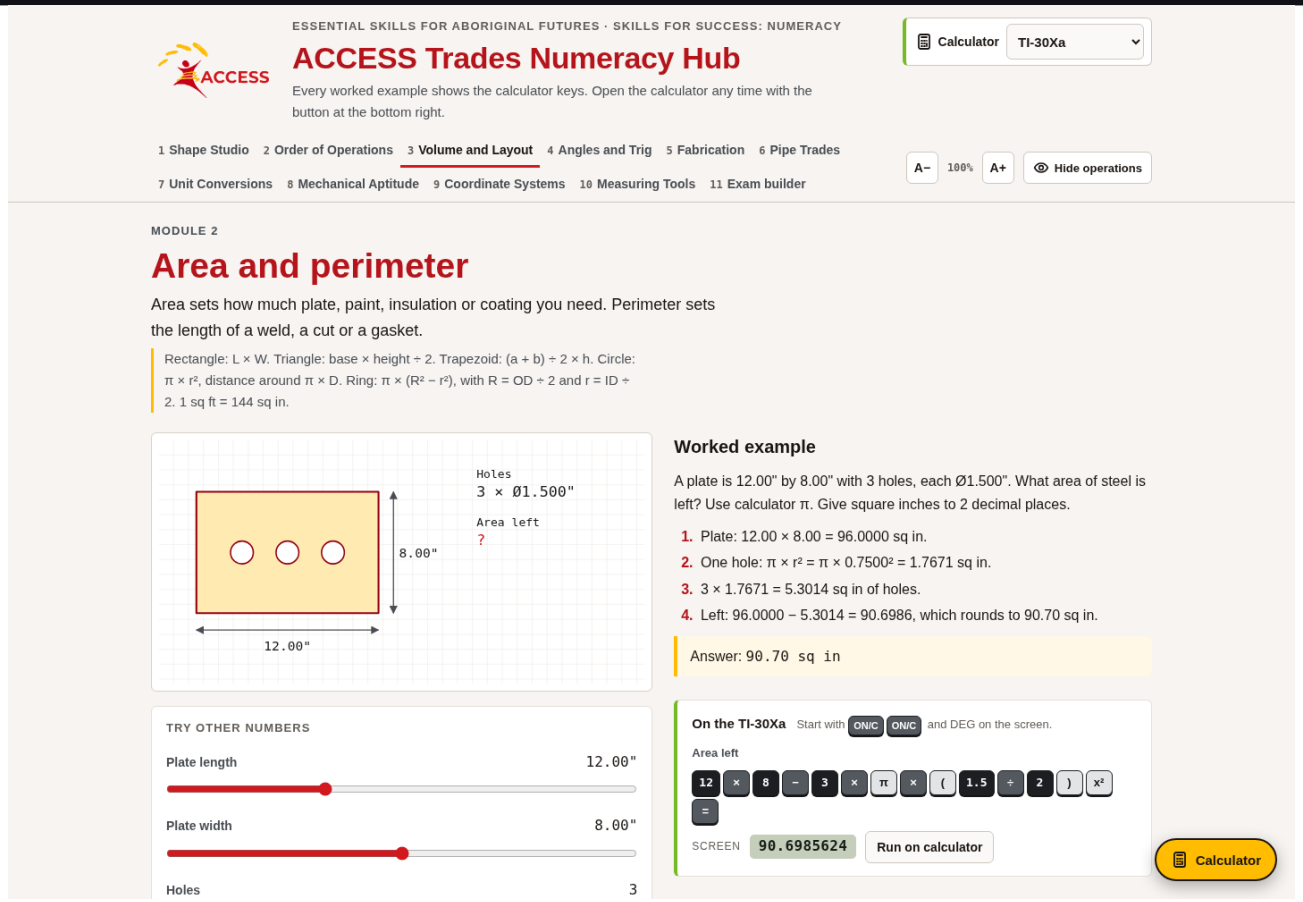
Complete Exercise Book 9A Q2–5, 9B Q1 and 9C Q1. Create a dimensioned frame on Shape Canvas; calculate by both addition and formula.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 9A Q1–8 (p.85); 9B Q1 (p.87); 9C Q1–2 (p.89)

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Practise in Numeracy Hub and Shape Canvas

1. Use Shape Canvas to inspect a rectangle and triangle; trace only their outside edges.

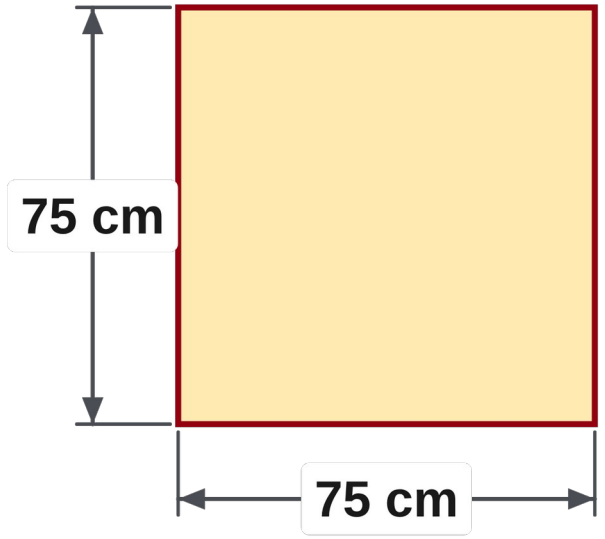
2. Create the assigned frame and calculate its perimeter by addition and formula.

3. Check the method in Numeracy Hub Volume and Layout → Area and perimeter.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Practise from the Exercise Book | 9A Q2

Find the perimeter of a square with sides of 75 cm.



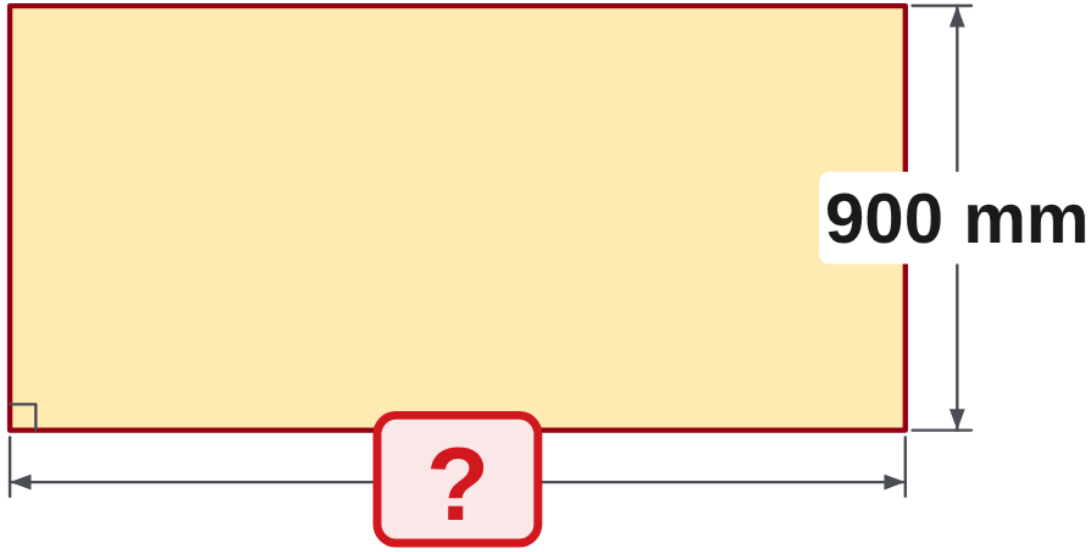
Perimeter ?

Write the operation, working and units before checking.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Practise from the Exercise Book | 9C Q1

The perimeter is 5,600 mm. Find the length.



Write the operation, working and units before checking.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Check independently

Exercise Book 9A Q8 and 9B Q1; explain the units and show the boundary calculation.

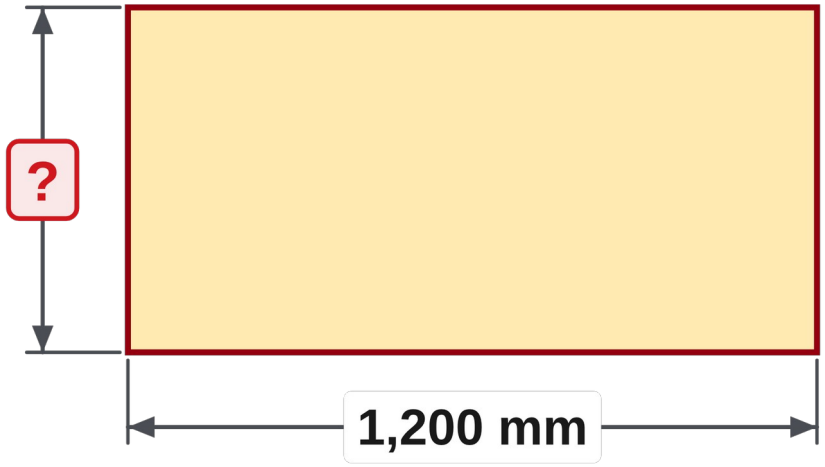
Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 9A Q1–8 (p.85); 9B Q1 (p.87); 9C Q1–2 (p.89)

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Independent check | 9A Q8

A rectangle has a perimeter of 3,800 mm and a length of 1,200 mm. Find the width.



Perimeter 3,800 mm

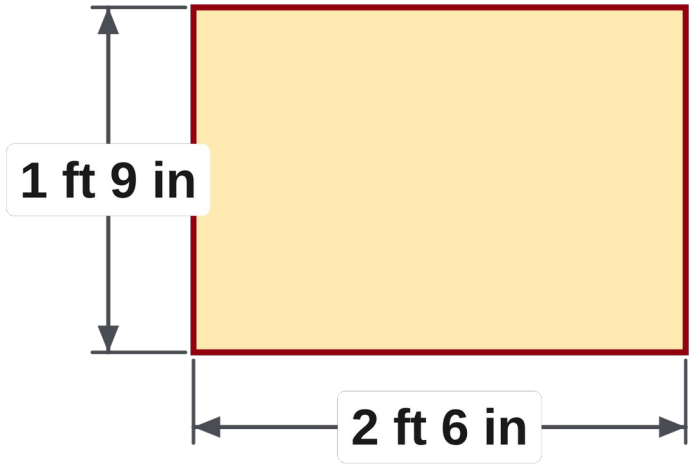
Width ?

Answer independently. Keep worked solutions closed.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Independent check | 9B Q1

A frame of angle iron is 2 ft 6 in by 1 ft 9 in. How much angle iron goes around it? Give feet and inches.



Angle iron around it ?

Answer independently. Keep worked solutions closed.

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 9A Q1–8 (p.85); 9B Q1 (p.87); 9C Q1–2 (p.89)

Textbook sections 9.1 pp.220–221; sections 9.6 pp.228–229

Break

10:15 AM–10:30 AM

Return ready for: Circumference, arcs and perimeter consolidation

This break is excluded from teaching minutes.

Circumference, arcs and perimeter consolidation

Today’s goal: Find circumference or an arc length and combine curved and straight boundary segments.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 9.2–9.6 pp.222–229

Exercise Book 9A Q9–17 (p.86), selected; 9B Q2,8 (p.87); 9C Q3–5 (pp.89–90)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 9.2–9.6 pp.222–229

Recall and reconnect

Try to label r and D on a circle and explain their relationship.

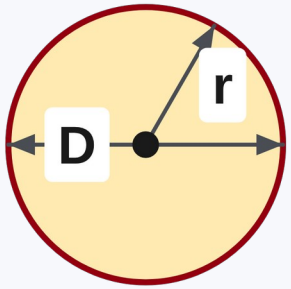
Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 9A Q9–17 (p.86), selected; 9B Q2,8 (p.87); 9C Q3–5 (pp.89–90)

Textbook sections 9.2–9.6 pp.222–229

Distinguish radius from diameter

1. Radius runs from the centre to the circle; diameter runs across through the centre.
2. D = 2r, so C = πD and C = 2πr describe the same circumference.
3. Use the π key and retain precision until the final answer.
4. Circumference is a length, so it uses a linear unit.



r: centre to edge

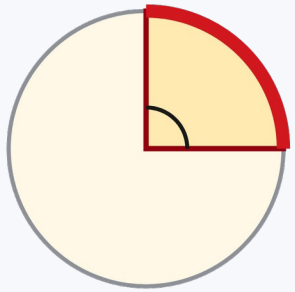
D: edge to edge through the centre

D = 2r

Textbook sections 9.2–9.6 pp.222–229

Combine curved and straight boundaries

1. A full circumference represents 360°.
2. Arc length = angle/360 × full circumference.
3. Add the straight boundary segments shown in the problem.
4. Draw the complete boundary first so no required edge is omitted or counted twice.



arc =

angle ÷ 360

× circumference



+ straight diameter

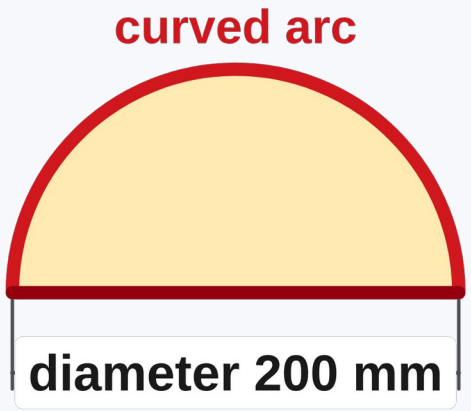
Textbook sections 9.2–9.6 pp.222–229

An arc is not always the whole perimeter

Classroom example • A circle has diameter 200 mm.
C = π × 200 ≈ 628.32 mm.

90° arc = ¼ × 200π ≈ 157.08 mm
Semicircular arc = ½ × 200π ≈ 314.16 mm
Semicircle perimeter = 100π + 200 ≈ 514.16 mm

Why the difference? The semicircle's complete boundary includes the diameter. The curved arc alone does not.



Together • Work 9A Q11–13,15. Sketch before calculating.
Shape Canvas: change a circle's radius and predict its circumference.

Classroom example, not a reproduction of an unseen book question • Numeracy Hub: Area and perimeter, Easy

Textbook | page 222

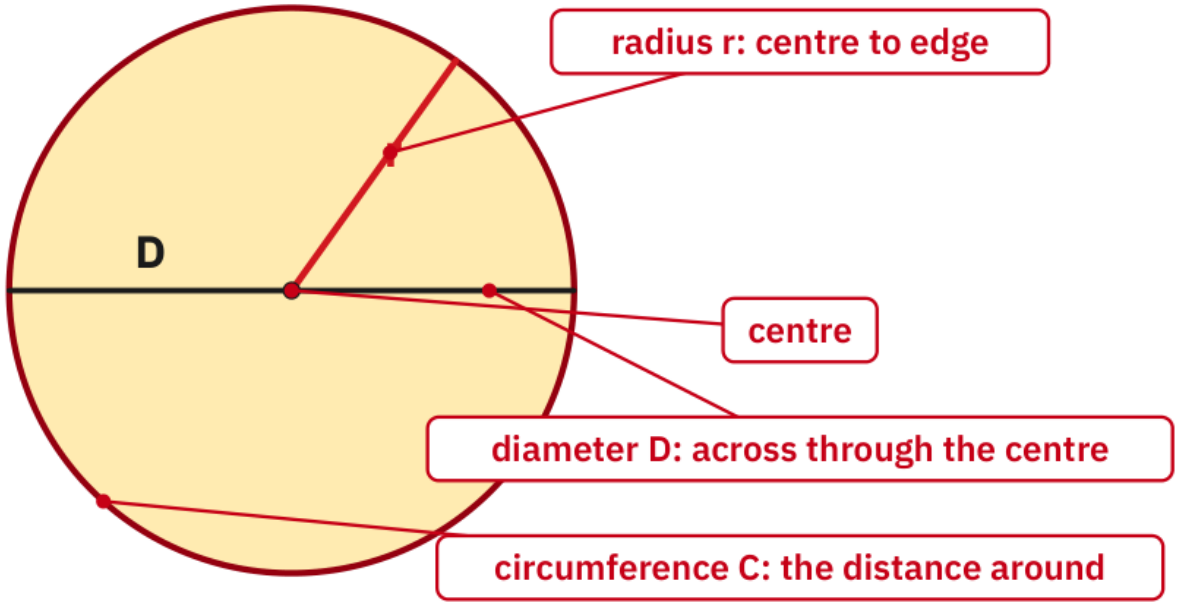


Figure 9.1 The parts of a circle.

Textbook p.222 • Figure 9.1 — Radius, diameter and circumference

Textbook sections 9.2–9.6 pp.222–229

Textbook | page 227

Example 9.9 A curved handrail

A curved handrail follows a 120° arc of a circle 2,400 mm in diameter. How long is the rail? Round to two decimal places.

SOLUTION

1. 120° is 120 ÷ 360 = one third of the circle.
2. arc = 120 ÷ 360 × π × 2,400 = 2,513.27

Textbook p.227 • Example 9.9 — A curved handrail as a fraction of a circumference

Textbook sections 9.2–9.6 pp.222–229

Spot the mistake. Explain the correction.

A semicircle’s perimeter is entered as half its circumference. What boundary is missing?

Discuss the method before checking the answer.

The straight diameter must also be included when the full semicircular boundary is required.

Textbook sections 9.2–9.6 pp.222–229

Work together

Work Exercise Book 9A Q11–13,15 together; sketch each circle or arc before calculating.

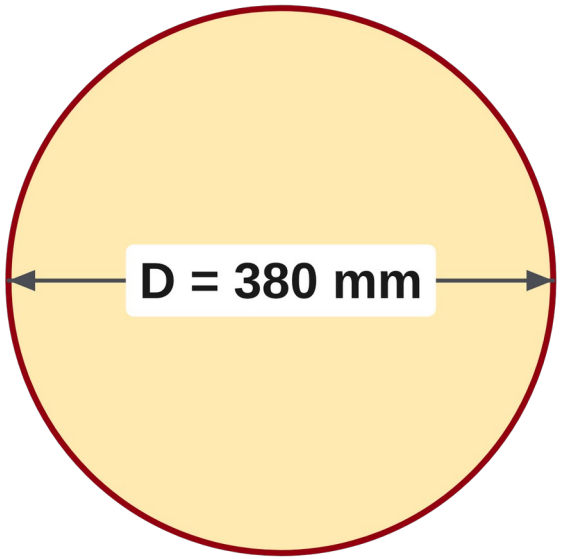
Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 9A Q9–17 (p.86), selected; 9B Q2,8 (p.87); 9C Q3–5 (pp.89–90)

Textbook sections 9.2–9.6 pp.222–229

Guided practice | 9A Q11

D = 380 mm. Find C to two decimal places.



Circumference C ?

Work with a partner. Take turns explaining the next step.

Textbook sections 9.2–9.6 pp.222–229

Compare your method | 9A Q11

D = 380 mm. Find C to two decimal places.

CHECK TOGETHER

1,193.81 mm

Textbook sections 9.2–9.6 pp.222–229

Your scheduled practice

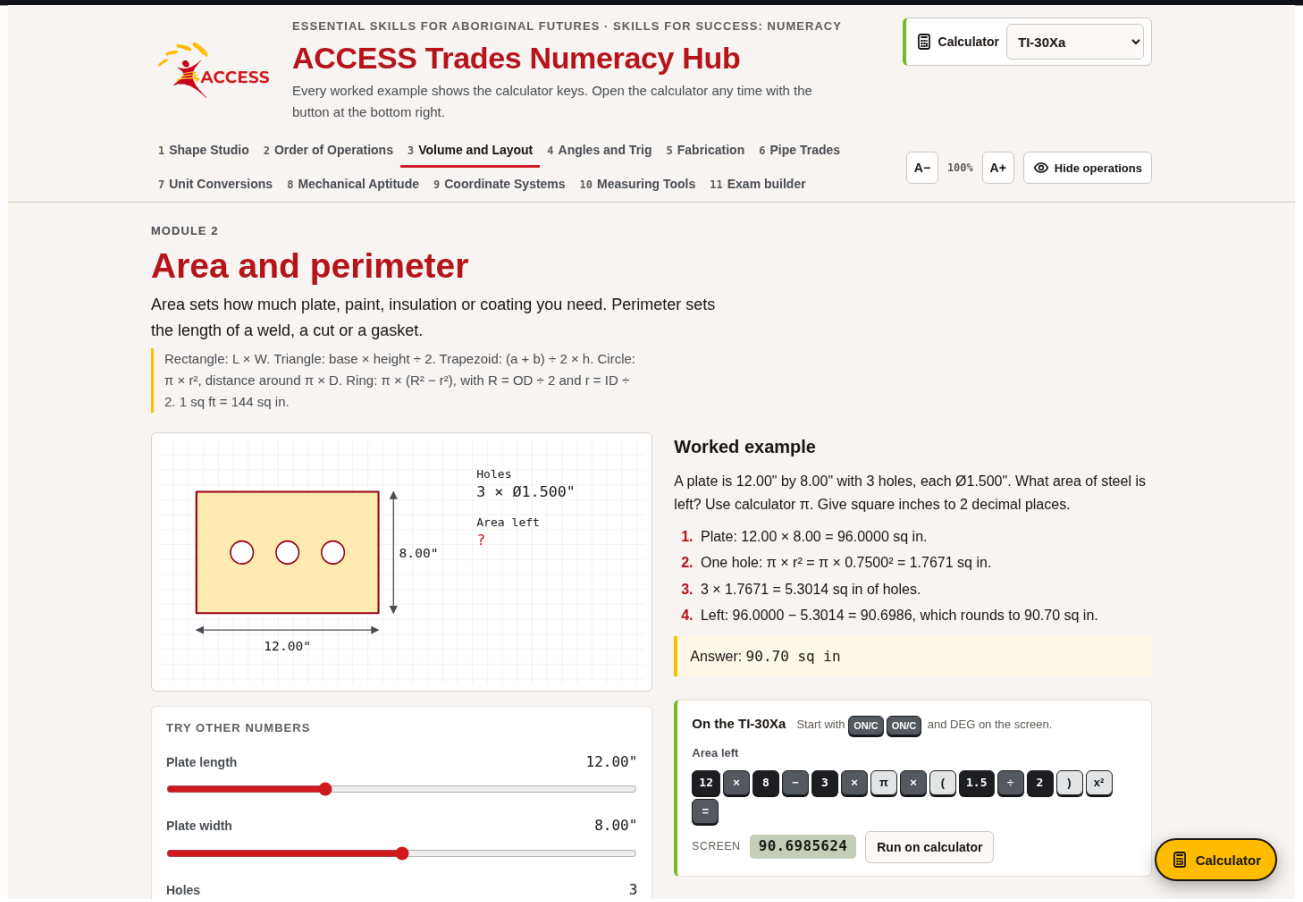
Solve Exercise Book 9B Q2,8 and 9C Q3–5. Use extra time to compare two methods and correct radius/diameter errors. Tool answers stay hidden until a paper attempt.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 9A Q9–17 (p.86), selected; 9B Q2,8 (p.87); 9C Q3–5 (pp.89–90)

Textbook sections 9.2–9.6 pp.222–229

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 9.2–9.6 pp.222–229

Practise in Numeracy Hub and Shape Canvas

1. In Shape Canvas, change a circle’s radius and predict how its circumference changes.

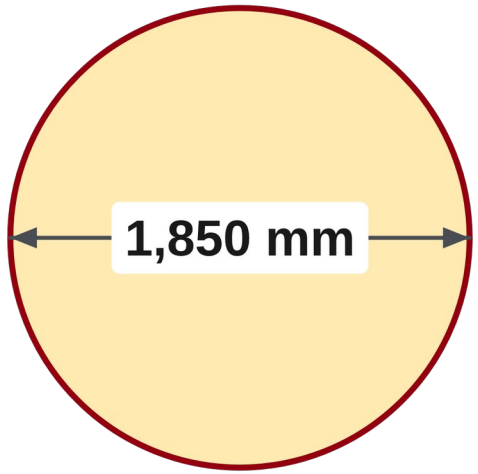
2. Sketch and calculate the assigned circle or arc on paper with r and D labelled.

3. Check an Easy Area and perimeter example in Numeracy Hub after the paper attempt.

Textbook sections 9.2–9.6 pp.222–229

Practise from the Exercise Book | 9B Q2

A steel band goes once around a tank 1,850 mm in diameter, with 150 mm extra for the overlap. Find the band length to the nearest millimetre.



Overlap 150 mm

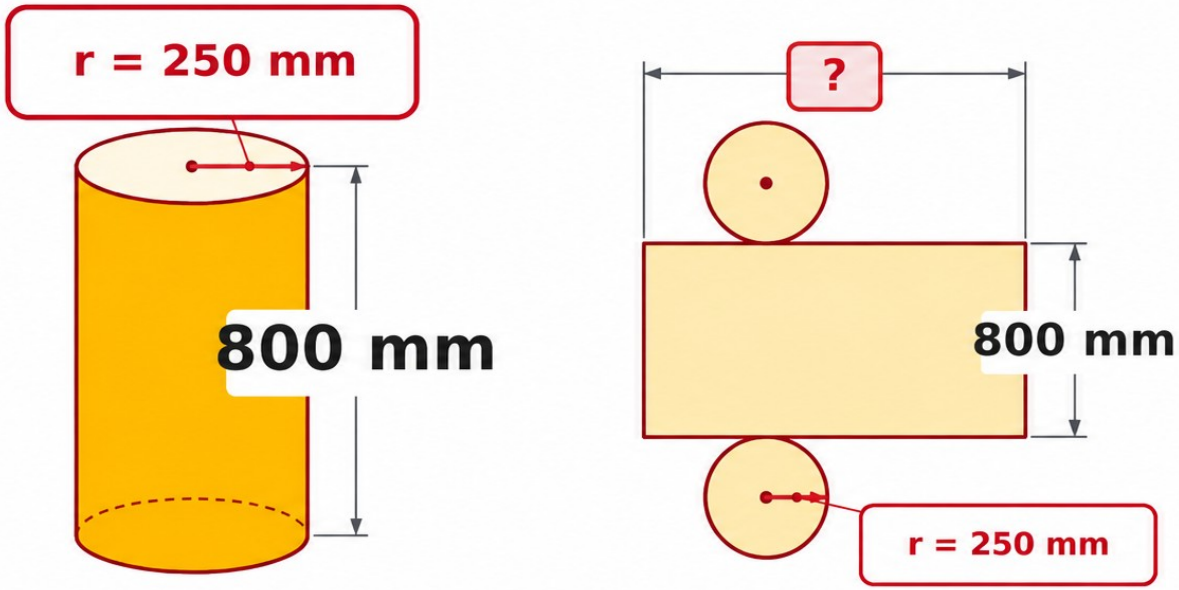
Band length ?

Write the operation, working and units before checking.

Textbook sections 9.2–9.6 pp.222–229

Practise from the Exercise Book | 9C Q3

Unfold it. Find the stretch-out marked ?.



Write the operation, working and units before checking.

Textbook sections 9.2–9.6 pp.222–229

Check independently

Exercise Book 9A Q11 and Q15; label diameter and identify the arc fraction.

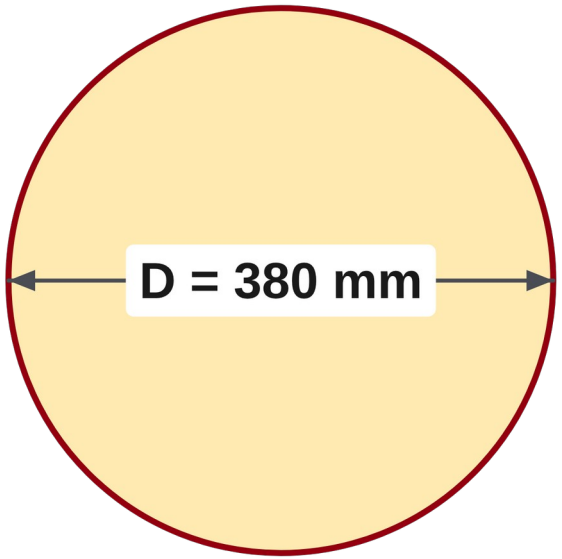
Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 9A Q9–17 (p.86), selected; 9B Q2,8 (p.87); 9C Q3–5 (pp.89–90)

Textbook sections 9.2–9.6 pp.222–229

Independent check | 9A Q11

D = 380 mm. Find C to two decimal places.



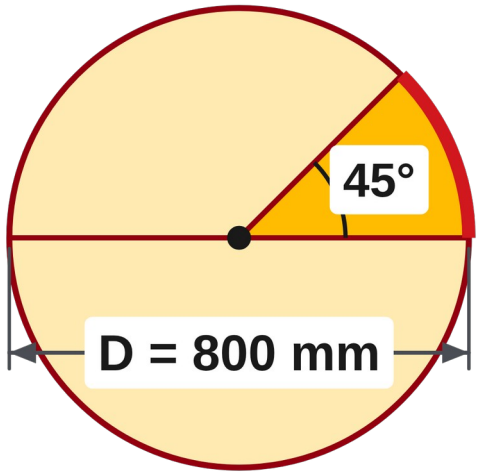
Circumference C ?

Answer independently. Keep worked solutions closed.

Textbook sections 9.2–9.6 pp.222–229

Independent check | 9A Q15

Find the length of a 45° arc of a circle with D = 800 mm. Round to two decimal places.



Arc length ?

Answer independently. Keep worked solutions closed.

Textbook sections 9.2–9.6 pp.222–229

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 9A Q9–17 (p.86), selected; 9B Q2,8 (p.87); 9C Q3–5 (pp.89–90)

Textbook sections 9.2–9.6 pp.222–229